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DNN - Phase 1 Report

1. Problem formulation

The goal is to train a model on Sign Language MNIST in order to classify sign language signs with 95% accuracy given correctly formatted input images.

Test data is manually collected and described in Section 4.

https://www.kaggle.com/datasets/datamunge/sign-language-mnist/data

2. Existing solutions

2.1. Convolutional Neural Networks (CNNs)

CNNs are the standard approach for image classification tasks like MNIST or sign language datasets. CNNs contain an input layer, hidden layers and an output layer. One of the hidden layers performs convolution (typically a dot product of a kernel and input).

2.1.1. LeNet

Link to the paper: https://ieeexplore.ieee.org/document/726791

LeNet (1998 version, also called LeNet-5) is the foundational CNN architecture. It was designed for handwritten digit recognition. It contains convolutional and pooling layers, followed by fully connected layers and was the basis for all the following CNN architectures.

2.1.2. Custom CNN, ResNet-50, EfficientNet-B0, and Inception-V3

Paper: https://doi.org/10.61173/yackf205

These architectures are much newer than LeNet. In the comparative study, the highest accuracy of 99.8% was achieved by EfficientNet-B0 out of all these architectures. It used various techniques to achieve the best models possible: data augmentation, transfer learning, and fine-tuning to the ASL Alphabet dataset.

2.2. Transformers

Link to the paper (*Attention Is All You Need*): https://arxiv.org/abs/1706.03762

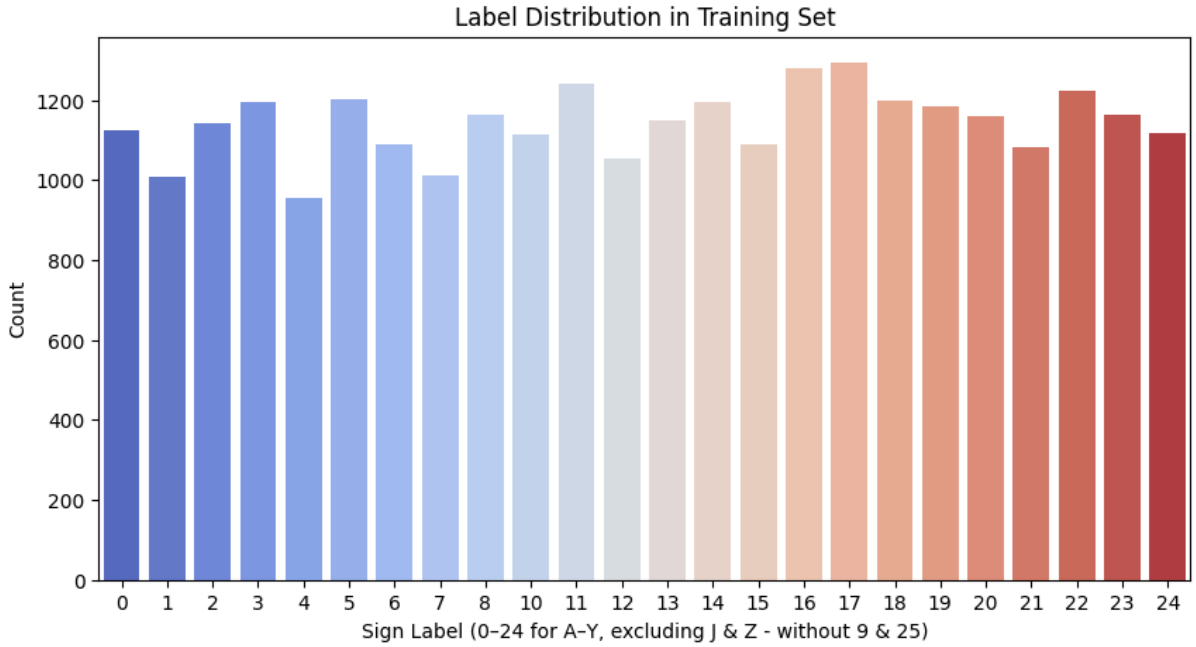
Transformers are deep neural architectures that use the concept called **attention**. In transformers, text is transformed into numerical tokens and vectors. The *attention* means that the most important tokens are given more “attention”.

3. Training data

We took train and test data provided on Kaggle and we treat test data as validation data instead (because test data is collected manually by us).

Here is the information about these files from Kaggle:

- 7172 rows (photos) in CSV file *sign_mnist_test.csv*.
- 27.5K rows (photos) in CSV file *sign_mnist_train.csv*.
- First column in CSV files describes the class of the photos, the rest of the columns describe the shade of gray of the pixel. Name of the column is the pixel number (ordered from top-left of the image and going right then bottom).
- 24 Classes (English alphabet without letters ‘j’ and ‘z’ due to movement needed to express those signs).



More detailed description (source: *Kaggle*):

The dataset format is patterned to match closely with the classic MNIST. Each training and test case represents a label (0-25) as a one-to-one map for each alphabetic letter A-Z (and no cases for 9=J or 25=Z because of gesture motions). The training data (27,455 cases) and test data (7172 cases) are approximately half the size of the standard MNIST but otherwise similar with a header row of label, pixel1,pixel2….pixel784 which represent a single 28x28 pixel image with grayscale values between 0-255. The

original hand gesture image data represented multiple users repeating the gesture against different backgrounds. The Sign Language MNIST data came from greatly extending the small number (1704) of the color images included as not cropped around the hand region of interest. To create new data, an image pipeline was used based on ImageMagick and included cropping to hands-only, gray-scaling, resizing, and then creating at least 50+ variations to enlarge the quantity

4. Test data

- Photos of sign language signs made with smartphone in natural light in front of a wall, to mimic real life usage.
- 24 photos corresponding to 24 classes, in future there is possibility to add next 24 photos done by a different person to add variety.
- 24 Classes (English alphabet without letters ‘j’ and ‘z’ due to movement connected with expressing those signs).
- Image processing will be done in Python.

Example image:

